

# MOHNISH KODUKULLA

Full-Stack & AI Systems Engineer | GSoC 2026 @ Debezium | DevOps  
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## EDUCATION

### B.Tech in Computer Science and Engineering

Vellore Institute of Technology, Chennai

Aug 2023 - Present

GPA: 9.28/10

- **Completed Coursework:** Database Management Systems, Artificial Intelligence, Data Structures and Algorithms, Operating Systems, Computer Networks, Probability & Statistics, Blockchain Technology

## SKILLS

**AI/ML Technologies:** LLMs, RAG, LangChain, LangGraph, Prompt Engineering, Agentic AI, n8n

**DevOps & Cloud:** Docker, Redis, Git, GitLab CI/CD, AWS EC2

**Frameworks:** React.js, Next.js, Tailwind CSS, Flask, Django, FastAPI

**Monitoring Systems:** Prometheus, Grafana, WebSockets, Distributed Systems

**Databases:** PostgreSQL, MongoDB, SQL

**Languages:** Python, JavaScript, C++, Java, SQL, HTML/CSS

## WORK EXPERIENCE

### Debezium

May 2026 - Present

GSoC Contributor – AI Infrastructure & Developer Tooling

Remote

- Building **PyDebeziumAI**, a CDC integration layer connecting Debezium change-data-capture streams to LangChain and LangGraph pipelines
- Designing semantic synchronization pipelines that propagate relational database changes to vector stores in near real-time, keeping embeddings consistent with source-of-truth data
- Authored Sphinx documentation across two merged PRs (multi-round review), fixing runtime bugs and adding nitpicky CI mode

### SRIP – VIT Chennai

May 2025 - Jul 2025

Summer Research Intern – AI Systems & Full-Stack Development

Chennai, India

- Built **HintGen** and **SALS**, AI-powered learning systems for contextual hint generation, personalized quizzes, and weak-topic analysis using RAG and evaluation pipelines
- Explored orchestration workflows, semantic retrieval systems, and scalable AI-assisted educational pipelines

## PROJECTS

### Hint Generation System (RAG-based)

[View Project](#)

- Built a contextual AI hint generation system for coding platforms using RAG workflows
- Designed a 3-stage pipeline for code evaluation, hint generation, and scoring using LLM orchestration
- Used RAG with TF-IDF similarity, achieving >95% relevant context retrieval
- Maintained response time of 15–20 seconds with optimized context handling and caching

### SALS – Smart Adaptive Learning System

[View Project](#)

- Developed an adaptive AI learning platform for personalized quiz generation and weak-topic analysis
- Improved accuracy of weak-topic detection by 30% and reduced manual effort

### Healthcare IoT Monitoring System

[View Project](#)

- Designed a microservices-based healthcare monitoring system for real-time patient vitals tracking
- Integrated observability pipelines using MQTT, Prometheus, and Grafana
- Implemented layered security with TLS, encryption, per-device keys, and JWT authentication

## OPEN SOURCE

### Debezium (CDC Platform)

[Pull Requests](#)

- Contributed to Debezium Core, Server, and Examples, focusing on MongoDB connector logic, schema handling, and CDC workflows

### OpenROAD (VLSI Physical Design Platform)

[Pull Requests](#)

- Merged fix in **dbSta** correctly propagating clock signal type when Liberty timing data is loaded after LEF
- Merged fix in **odb** adding validation for 3D block-vector region coordinate counts

## ACHIEVEMENTS

- **Google Summer of Code 2026 Contributor @ Debezium**
- **Runner-Up – ILH Hackathon** for an AI-driven blockchain stock market platform
- Solved 275+ DSA problems on LeetCode

## LEADERSHIP

### Development Lead, CYSCOM VIT Chennai

Aug 2024 – Present

- Led AI-driven projects and mentored juniors in full-stack development, system design, and engineering workflows